

PROJECT PROPOSAL

Modeling and Simulation

I. Title

Optimizing Multi-Channel Order Fulfillment and Queue Dynamics at Carlo's Bacolod Chicken House Express (CHE) Makati: A Discrete-Event Simulation Approach

II. Introduction

What the project is about

Carlo's Bacolod Chicken House Express (CHE), located at Savana Market along Chino Roces Avenue in Makati City, is a high-volume restaurant known for authentic Bacolod-style chicken inasal. During its peak lunch (11:30 AM – 1:30 PM) and dinner (6:00 PM – 8:30 PM) hours, CHE serves a simultaneous mix of dine-in patrons, walk-in takeout customers, and third-party delivery riders (e.g., GrabFood and Foodpanda).

This project builds a discrete-event simulation (DES) of CHE's order-to-dispatch process using Python (SimPy). Customer arrivals, cashiering, and charcoal-grilling times are treated as random (stochastic) processes whose distributions are fitted from field observations. The model then serves as an experimental testbed for staffing, queue-layout, and dispatch decisions.

Why it is important

Unlike a single-counter cafeteria, CHE operates as a coupled multi-stage system: (1) order taking and cashiering, (2) charcoal-grilling batch production, and (3) order assembly and dispatch (Figure 1). When arrivals surge, a delay at one stage backs up into the stages before it. The result is long lines at the entrance and counter, aisle and sidewalk congestion in the Savana Market corridor, slower table turnover, and customers who leave without ordering (balking) once the crowd looks too long. Each of these costs the restaurant sales and goodwill and leaves delivery riders waiting.

What problem it addresses

Trying out changes directly in a busy, live restaurant is risky and expensive. Simulation lets the team compare different staffing allocations, order-staging setups, and dispatch priorities on a computer first, without disturbing service or kitchen operations, so that only the best-performing changes are proposed to management.

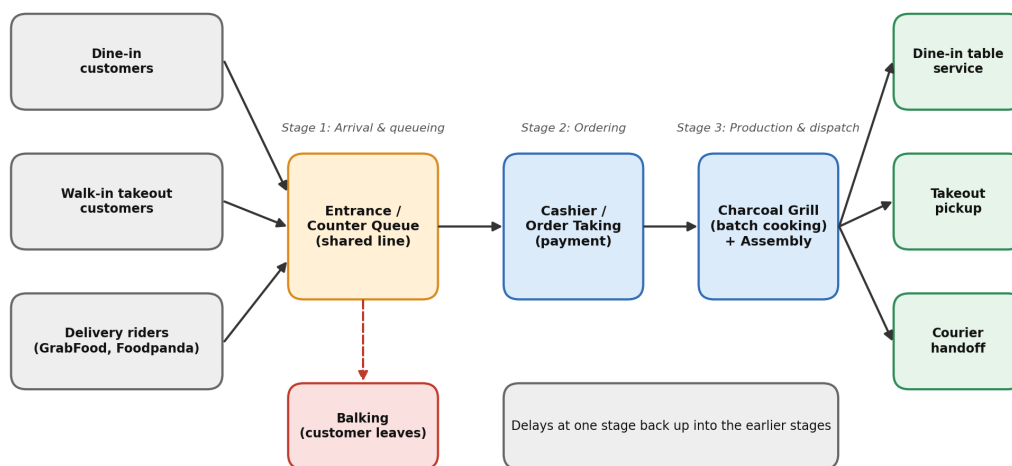


Figure 1. Simplified order-fulfillment flow at CHE

III. Problem Statement

During stochastic lunch and dinner rushes, long customer and courier waiting times and crowded queues at CHE reduce service quality and lead to lost sales. The central question of this study is:

How can Carlo's Bacolod Chicken House Express (CHE) reduce customer waiting time (W_q), shorten physical queue length (L_q), and minimize order fulfillment delays during peak hours through better order staging, line segmentation, and staff allocation, while keeping staff and grilling-station utilization (ρ) at balanced levels?

The study will judge performance using the measures in Table 1.

Table 1. Key performance measures

Measure	Symbol	Meaning
Average waiting time in queue	W_q	Average time a customer or rider waits before being served at the counter
Average queue length	L_q	Average (and maximum) number of people waiting in line
Utilization	ρ	Share of time a cashier or grill/dispatch station is busy; a low value signals idle time, a value near 1 signals overload
Order fulfillment time	—	Time from payment to food handover
Courier dwell time	—	Time from a rider's arrival to departure with the order
Balking rate	—	Share of arriving customers who leave without ordering
Throughput	—	Number of orders completed per hour

IV. Objectives

General Objective

To model, analyze, and optimize the queue dynamics, service flow, and multi-channel fulfillment efficiency of Carlo's Bacolod Chicken House Express (CHE) in Makati during peak hours using a discrete-event simulation approach.

Specific Objectives

1. **Collect data.** Gather field observations during 3–5 peak lunch sessions to measure inter-arrival times (dine-in, takeout, and delivery), cashier transaction times, payment-to-handover times, queue lengths, and balking incidents. The target is at least 100 inter-arrival observations per channel and at least 50 service-time observations per station, where volume allows.
2. **Analyze the system.** Map CHE's service paths, identify the bottleneck stage(s), and fit the collected data to statistical distributions using goodness-of-fit tests.
3. **Develop the model.** Build a baseline DES model of CHE's current workflow in Python (SimPy) and validate it so that its simulated averages of W_q , L_q , and throughput are within 10% of the observed values (or fall within the 95% confidence interval of the simulation).
4. **Test and recommend.** Run what-if scenarios (decoupling takeout/delivery from dine-in, adding a floating order-taker, and dynamic peak staffing) and recommend the configuration that minimizes customer and courier wait times while keeping staff utilization balanced and sustainable.

V. Scope and Limitations

Scope

- **Location:** Carlo's Bacolod Chicken House Express (CHE), Savana Commercial Center, 114 K Savana Market, Metropolitan Ave. cor. Chino Roces Ave., Brgy. Sta. Cruz, Makati City.
- **Customer channels:** Walk-in dine-in customers, walk-in takeout customers, and third-party delivery riders arriving at the order/counter area.
- **Time windows:** Peak lunch (11:30 AM – 1:30 PM) on observed days.
- **System stages modeled:** Arrival and queueing, ordering and payment, grilling and order assembly, and dispatch/handoff.
- **Modeling tool:** Discrete-event simulation in Python (SimPy), with data analysis in pandas and scipy.stats (AnyLogic as an alternative).

Limitations

- **Kitchen internals:** Charcoal-grilling batch cycles are modeled as random service delays, not as the heat transfer or cooking physics of the grill.

- **Seating and floor movement:** The model focuses on queue formation, order placement, and dispatch handoff. Table-holding time is a bounded service delay, and detailed floor-plan pathfinding is not covered.
- **Observation only:** Data are gathered by stopwatch and tally from public customer areas, without interviewing or interfering with staff. Internal data such as sales records, inventory, or staff schedules are not used.
- **Limited sample:** Results rely on 3–5 observed sessions, so the fitted distributions represent typical peak conditions and may not capture unusual days (holidays, promotions, bad weather).
- **Outside the study:** Off-peak hours, cost and profit analysis, menu changes, and actual implementation of the recommendations.

VI. Methodology

The project follows the seven steps shown in Figure 2.

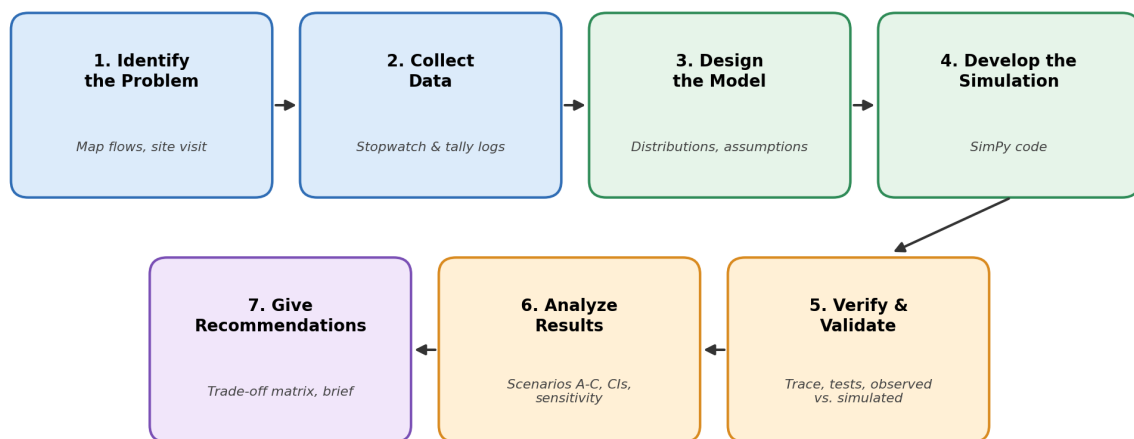


Figure 2. Methodology flow of the study

Step 1. Identify the Problem

The team will visit the site during off-peak hours to sketch CHE's layout, trace the paths of dine-in, takeout, and delivery customers, and list where waiting occurs. The result is a flow map (arrival → queue → cashier/order → waiting area → order dispatch/fulfillment) and a list of the performance measures in Table 1.

Step 2. Collect Data

Non-intrusive stopwatch and tally observations will be made over 3–5 peak lunch and dinner sessions, recorded in 15-minute bins so that changes in arrival intensity during the rush can be captured. No personal information about customers or staff will be recorded, and the team will inform or seek permission from CHE management before observing.

Table 2. Data to be collected

Variable	What is recorded	Use in the model
Inter-arrival times	Timestamp of each arrival, by channel (dine-in, takeout, delivery)	Arrival process for each channel
Cashier service time	Start of ordering to completion of payment	Service-time distribution at the counter
Order fulfillment time	Payment to food handover	Grill/assembly delay distribution
Courier dwell time	Rider arrival to departure	Validation measure and scenario output
Queue length	Head count at fixed intervals (e.g., every 5 minutes)	Validation of L_q
Balking incidents	Tally of people who join, look at, then leave the line	Balking rule and validation

Step 3. Design the Model

- **Distribution fitting:** Use `scipy.stats` to fit the data to candidate distributions (Exponential or Gamma for inter-arrival times; Lognormal, Gamma, or Triangular for service and fulfillment times), and select the best fit using Kolmogorov–Smirnov or chi-square goodness-of-fit tests plus Q–Q plots.
- **Time-varying arrivals:** Arrival rates will be estimated per 15-minute bin so the surge pattern of the rush is reproduced.
- **Entities and resources:** Customers, riders, and orders are the entities. Cashiers, the grill, and assembly/dispatch staff are the resources. Balking is modeled with a rule based on the observed queue length.
- **Key assumptions:** First-come, first-served discipline within a line; independent arrivals; and service times independent of queue length. These will be revisited after data collection.

Step 4. Develop the Simulation

The baseline multi-channel queueing system will be coded as a modular SimPy program with separate components for arrivals, the counter, the kitchen, and dispatch, so that scenarios can be switched on by changing parameters rather than rewriting code. Random-number streams will be seeded for reproducibility.

Step 5. Verify and Validate

- **Verification (is the model built right?):** Trace sample customers step by step through the event log, test extreme cases (no arrivals, a single very slow cashier), unit-test the code, and confirm the outputs satisfy Little’s Law ($L = \lambda W$).
- **Validation (is it the right model?):** Compare simulated W_q , L_q , throughput, and balking rate against the observed values, preferably from a session not used for fitting. The baseline is accepted if simulated means fall within 10% of observed means or the

observed values fall inside the simulation's 95% confidence interval. Where possible, the process map will also be checked with a CHE staff member (face validation).

Step 6. Analyze Results

Each scenario in Table 3 will be run for 30 independent replications covering a full peak window, using common random numbers so that differences reflect the policy and not random luck. Results will be reported as means with 95% confidence intervals and compared using paired differences. A sensitivity analysis will test how each scenario performs when arrival rates are raised or lowered by 20%.

Table 3. Scenarios to be simulated

Scenario	Description	Main measures of interest
Baseline	Current shared-counter, single-queue setup for dine-in, takeout, and delivery	W_q , L_q , ρ , balking rate
A. Channel decoupling	Separate ordering for dine-in from pickup for takeout and delivery, reducing queue interference and sidewalk crowding	Courier dwell time, W_q by channel
B. Floating expeditor	A roving staff member takes orders by mobile or paper slip during queue spikes, before customers reach the register	Peak L_q , balking rate
C. Dynamic peak staffing	One kitchen prep staff member is reassigned to order dispatch during the peak 60-minute surge	Fulfillment time, ρ of each station

Step 7. Give Recommendations

A scenario trade-off matrix and Pareto plots (waiting time versus staff utilization) will be used to select the most cost-effective changes. The recommendation brief will favor zero-to-low-cost layout and queue-management actions suited to CHE's Savana Market location.

VII. Expected Output

- **Validated simulation model:** A clean, modular Python (SimPy) program representing CHE's order-to-dispatch operations, with documentation on how to run each scenario.
- **Input-data package:** The cleaned observation dataset and a summary of the fitted distributions.
- **Scenario trade-off matrix:** A statistical comparison of average wait time (W_q), maximum queue length (L_q), courier dwell time, balking rate, and staff utilization across the baseline and alternative scenarios, with Pareto plots.
- **Operational recommendation brief:** Practical, zero-to-low-cost layout and queue-management recommendations for CHE's Savana Market location.
- **Research report and presentation:** The final written report and defense presentation.

VIII. Schedule of Activities

Table 4. Project timeline

Activity	Timeline	Key output
Proposal finalization and site reconnaissance	Week 8	Approved proposal, flow map
Field data collection and distribution fitting	Week 8	Observation dataset, fitted distributions
Baseline model coding and validation	Week 9	Validated baseline SimPy model
Scenario testing and sensitivity analysis	Week 9	Scenario trade-off matrix
Final report compilation and defense presentation	Week 10	Final report, slides, recommendation brief

IX. Resources Needed

- **Laptop/Computer:** Group members' personal laptops for data logging, statistical fitting, and running simulation replications.
- **Internet connection:** For research, cloud-based shared files, and team coordination.
- **Software tools:** Python 3 with SimPy, scipy, numpy, pandas, and matplotlib (or AnyLogic Personal Learning Edition); Google Sheets or Excel for observation logs and tally templates.
- **Data sources:** Primary field observations at CHE (stopwatch timestamps, head counts, and tallies on clipboard observation sheets), together with the recent (2020 onward) journal articles and conference papers listed in the References.
- **Field equipment:** Digital stopwatches or smartphones, tally counter apps, and printed observation sheets.

X. Expected Benefits

- **For customers and delivery riders (users):** Shorter standing time in line, faster food handover, and less crowding in the Savana Market corridor.
- **For CHE management (organization):** Evidence-based ways to relieve choke points, shorten rider dwell time, and reduce customer walkaways during peak hours without costly kitchen redesigns.
- **For the school/course:** A real-world case study demonstrating how simulation concepts (random processes, queueing, and model validation) apply to a Metro Manila business.
- **For the student researchers:** Hands-on experience in field data collection, statistical fitting, simulation programming, and communicating results to a decision-maker.

XI. Conclusion

Carlo's Bacolod Chicken House Express (CHE) offers a rich, high-volume setting with random multi-channel demand from dine-in, takeout, and delivery customers. By modeling its service pipeline with discrete-event simulation, this study goes beyond observation and provides a risk-free digital sandbox for testing improvements. The expected contribution is a validated model and a set of practical, low-cost recommendations that balance shorter customer and courier waiting times with sustainable staff workloads.

XII. References

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